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$f(x, y) = x^2 - y^2$

$$\begin{cases} \dfrac{\partial f}{\partial x} = 2x = 0 \\ \dfrac{\partial f}{\partial y} = -2y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$\Rightarrow (x, y) \in \{(0, 0)\}$

$$\begin{cases} \dfrac{\partial^2 f}{\partial x^2} = 2 \\ \dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial y \partial x} = 0 \\ \dfrac{\partial^2 f}{\partial y^2} = -2 \end{cases} \Rightarrow H(x, y) = -4$$

- $H(0, 0) = -4 < 0 \Rightarrow (0, 0)$ zadelpunt

References